APLIKASI UMKM RUPA REMPAH (SKINCARE)

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Tugas diajukan sebagai salah satu syarat
Untuk menyelesaikan Mata Kuliah Pemrograman Perangkat Bergerak

PROGRAM STUDI TEKNIK INFORMATIKA
FAKULTAS ILMU KOMPUTER
UNIVERSITAS DIAN NUSWANTORO SEMARANG
2026

Daftar Isi

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1. Latar Belakang

UMKM Rupa Rempah merupakan UMKM yang bergerak di bidang penjualan produk kecantikan dan perawatan kulit (Skincare). Untuk memperluas jangkauan penjualan dan mempermudah transaksi pelanggan, dikembangkanlah sebuah aplikasi katalog dan kasir berbasis mobile. Aplikasi ini diharapkan mampu mempermudah proses bisnis dan meningkatkan pengalaman belanja konsumen ke ranah digital.

2. Analisa dan Desain Aplikasi (OOP), Pemodelan UML

Aplikasi ini dikembangkan dengan konsep Object-Oriented Programming (OOP) menggunakan bahasa Dart dan framework Flutter. Struktur desain memisahkan antarmuka (UI) dan logika state management.

3. Analisa Kebutuhan User

- User dapat melakukan login ke dalam aplikasi menggunakan username dan password (SharedPreferences).
- User dapat melihat katalog produk Skincare berupa gambar, nama, dan harga.
- User dapat melihat deskripsi detail dari setiap produk.
- User dapat menambahkan produk ke dalam keranjang, menambah/mengurangi jumlah (qty), dan melihat total harga.
- User dapat melakukan proses pembayaran (checkout) dengan melihat struk belanja dan mendapatkan perhitungan kembalian.
- User dapat memperbarui data akun (User & Password).
- User dapat menggunakan pintasan call center, SMS, dan Maps penjual.

4. Analisa Kebutuhan Hardware dan Software

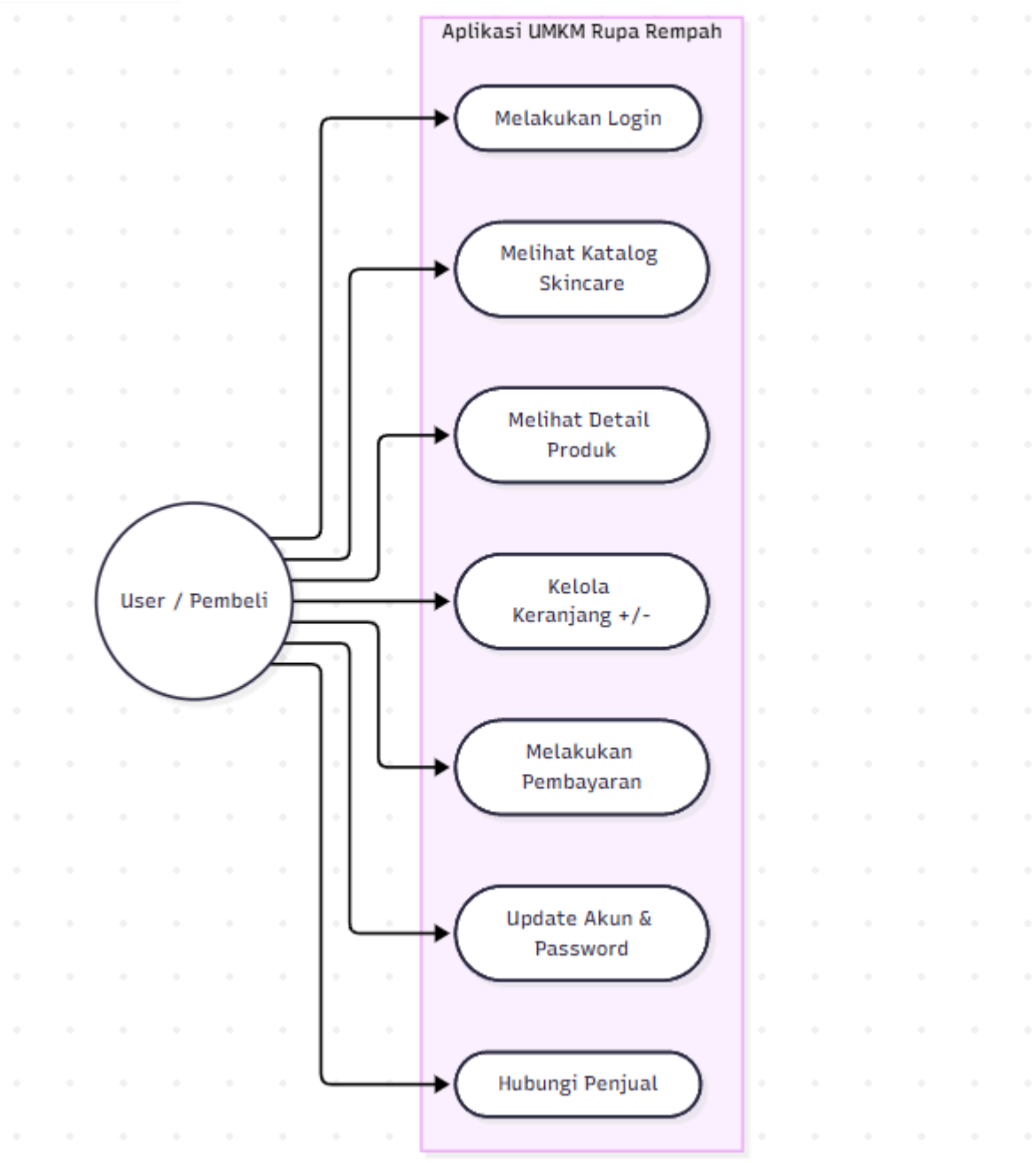
Kebutuhan Hardware:

- Personal Computer (PC) / Laptop untuk pengembangan.
- Smartphone Android / Web Browser (sebagai target implementasi).

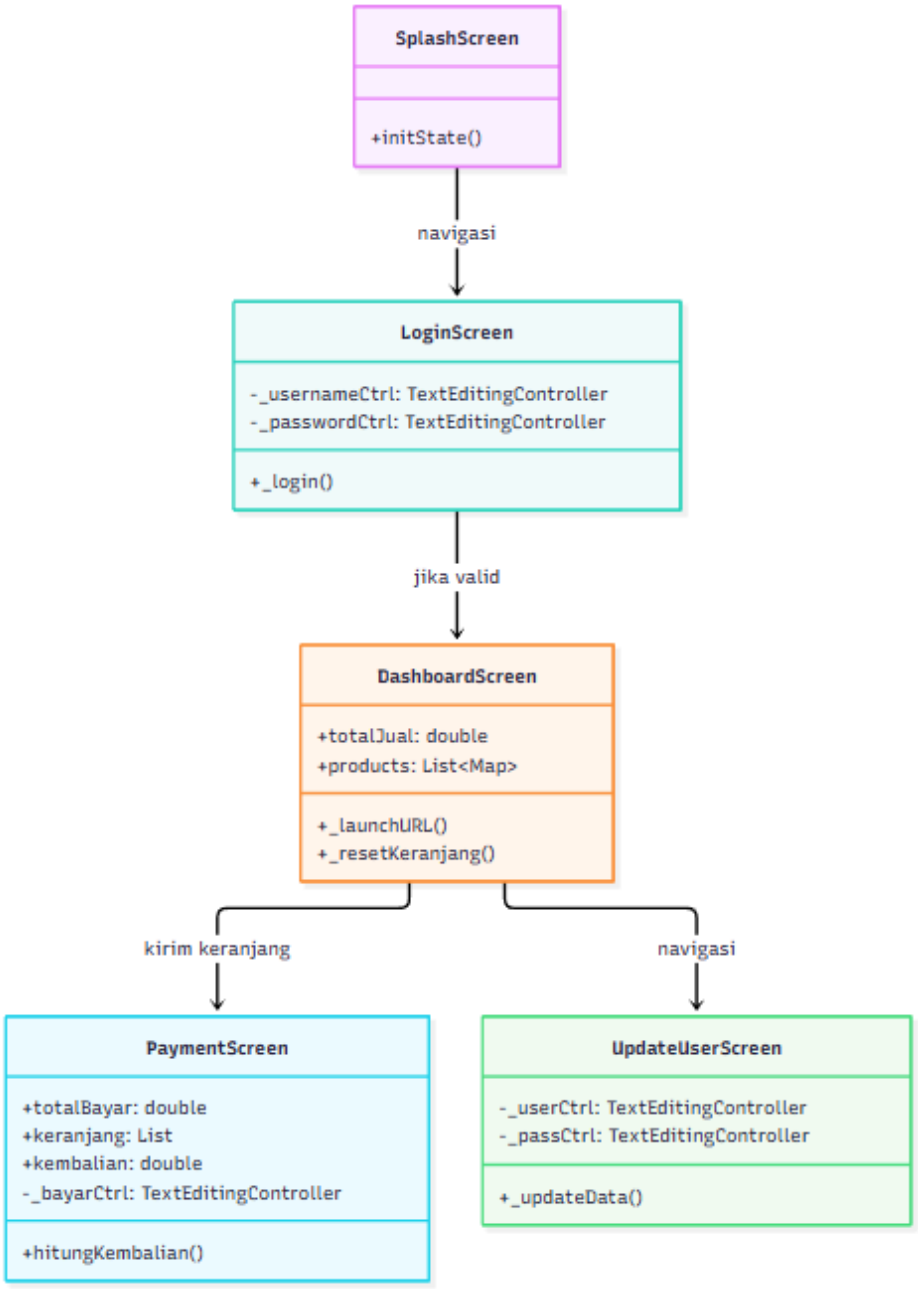
Kebutuhan Software:

- Sistem Operasi: Windows 10/11
- IDE: Visual Studio Code
- Framework: Flutter SDK
- Bahasa Pemrograman: Dart
- Version Control: Git & GitHub

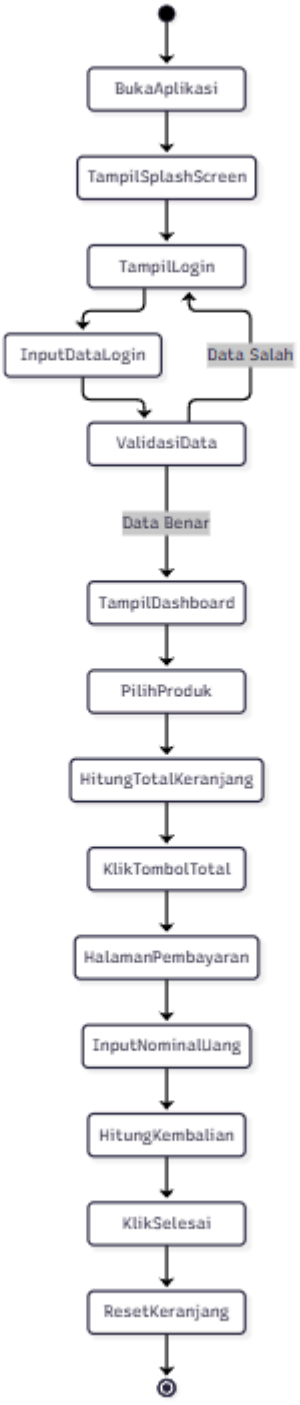
5. Diagram Use Case



6. Diagram Class



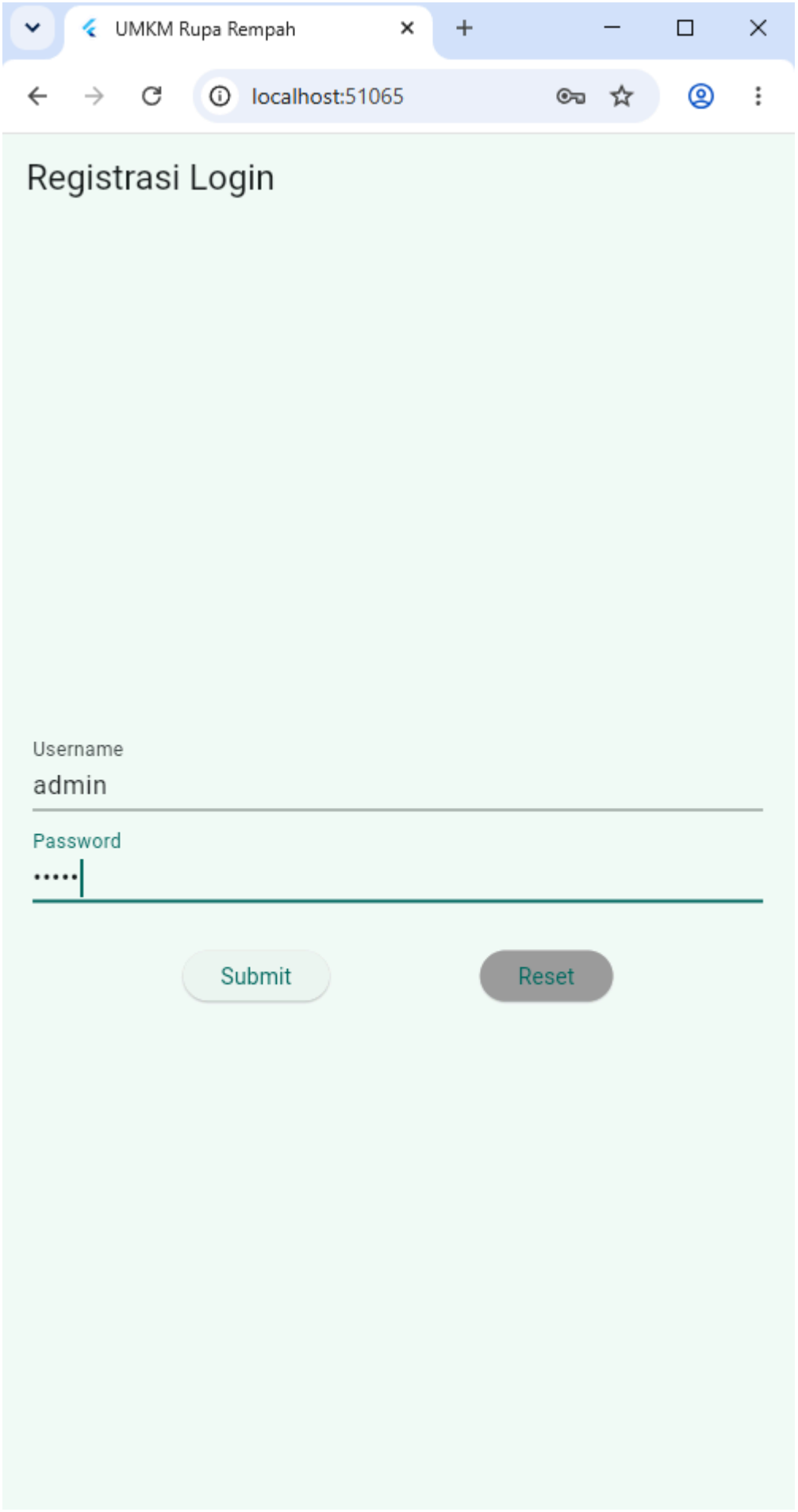
7. Diagram Aktifitas



8. Desain Input dan Output

Input: Form registrasi/login (username, password), tombol aksi produk (+/-), dan input form nominal pembayaran.

Output: Tampilan GridView produk, dialog deskripsi produk, struk daftar belanja pada halaman pembayaran, dan notifikasi kembalian.


UMKM Rupa Rempah
localhost:51065
Registrasi Login
Username
admin
Password
Submit
Reset

UMKM Rupa Rempah

localhost:51065

Katalog Produk



COSRX Set

Rp 150000

1



Lacoco Serum

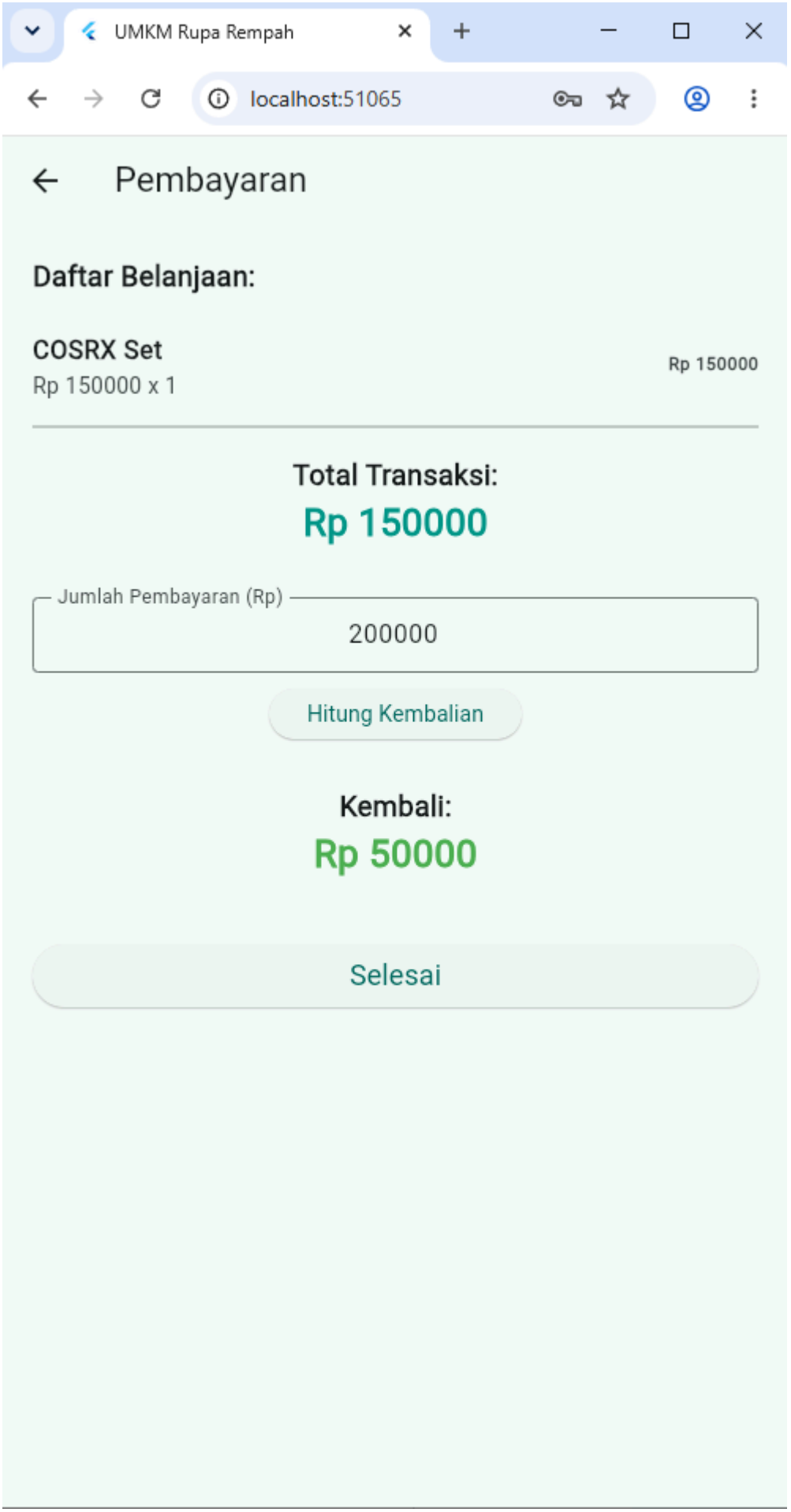
Rp 250000

0

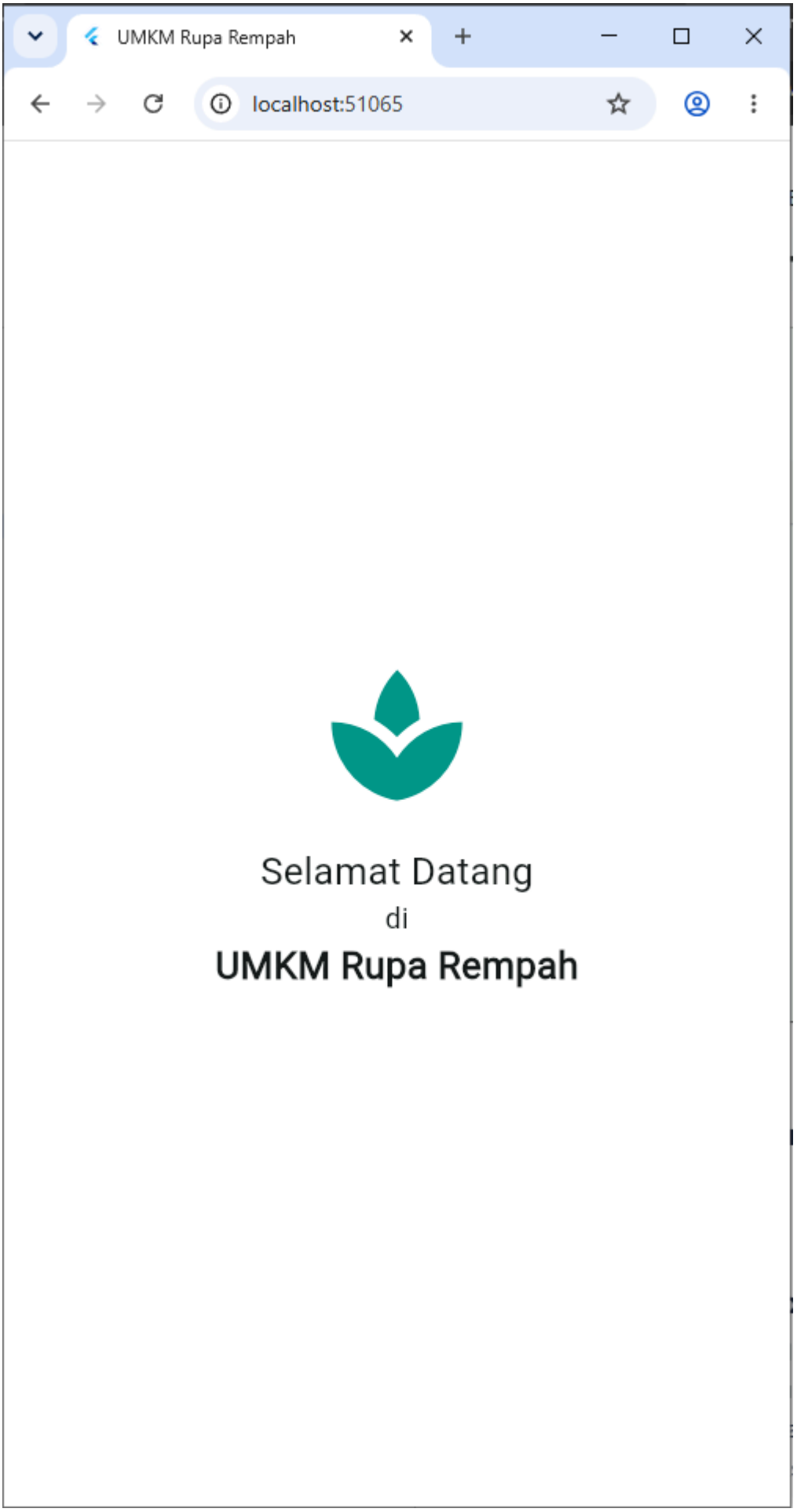




Total : Rp 150000


UMKM Rupa Rempah
localhost:51065
Pembayaran
Daftar Belanjaan:
COSRX Set
Rp 150000 x 1
Rp 150000
Total Transaksi:
Rp 150000
Jumlah Pembayaran (Rp)
200000
Hitung Kembalian
Kembali:
Rp 50000
Selesai

9. Implementasi Sistem


UMKM Rupa Rempah
localhost:51065
Selamat Datang
di
UMKM Rupa Rempah

10. Penutup

Pembuatan aplikasi mobile untuk UMKM Rupa Rempah telah berhasil diimplementasikan sesuai dengan kebutuhan sistem. Aplikasi telah mencakup seluruh alur utama mulai dari halaman autentikasi lokal (SharedPreferences), dashboard katalog skincare, manajemen keranjang interaktif, hingga transaksi kasir dan pembayaran.

11. Lampiran Code Program

main.dart

lib > main.dart > _SplashScreenState > initState

```dart
import 'package:flutter/material.dart';
import 'package:shared_preferences/shared_preferences.dart';
import 'package:url_launcher/url_launcher.dart';
import 'dart:async';

Run | Debug | Profile
void main() {
  runApp(const UtsApp());
}

class UtsApp extends StatelessWidget {
  const UtsApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'UMKM Rupa Rempah',
      theme: ThemeData(
        primarySwatch: Colors.teal,
        colorScheme: ColorScheme.fromSeed(seedColor: Colors.teal),
      ), // ThemeData
      debugShowCheckedModeBanner: false,
      home: const SplashScreen(),
    ); // MaterialApp
  }
}

// ========================================
// 1. SPLASH SCREEN
// ========================================
class SplashScreen extends StatefulWidget {
  const SplashScreen({super.key});

  @override
  State<SplashScreen> createState() => _SplashScreenState();
}

class _SplashScreenState extends State<SplashScreen> {
  @override
  void initState() {
    super.initState();
    Timer(const Duration(seconds: 3), () {
      Navigator.pushReplacement(
        context,
        MaterialPageRoute(builder: (context) => const LoginScreen()),
      );
```

```dart
    }); // Timer
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      backgroundColor: Colors.white,
      body: Center(
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: const [
            Icon(Icons.spa, size: 100, color: Colors.teal),
            SizedBox(height: 20),
            Text("Selamat Datang", style: TextStyle(fontSize: 24)),
            Text("di", style: TextStyle(fontSize: 18)),
            Text(
              "UMKM Rupa Rempah",
              style: TextStyle(fontSize: 24, fontWeight: FontWeight.bold),
            ), // Text
          ],
        ), // Column
      ), // Center
    ); // Scaffold
  }
}

// ========================================
// 2. LOGIN USER (Shared Preferences)
// ========================================
class LoginScreen extends StatefulWidget {
  const LoginScreen({super.key});

  @override
  State<LoginScreen> createState() => _LoginScreenState();
}

class _LoginScreenState extends State<LoginScreen> {
  final _usernameCtrl = TextEditingController();
  final _passwordCtrl = TextEditingController();

  Future<void> _login() async {
    final prefs = await SharedPreferences.getInstance();
```

```dart
    String savedUser = prefs.getString('username') ?? 'admin';
    String savedPass = prefs.getString('password') ?? 'admin';

    if (_usernameCtrl.text == savedUser && _passwordCtrl.text == savedPass) {
      Navigator.pushReplacement(
        context,
        MaterialPageRoute(builder: (context) => const DashboardScreen()),
      );
    } else {
      ScaffoldMessenger.of(context).showSnackBar(
        const SnackBar(content: Text('Username atau Password salah!')),
      );
    }
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('Registrasi Login')),
      body: Padding(
        padding: const EdgeInsets.all(20.0),
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            TextField(
              controller: _usernameCtrl,
              decoration: const InputDecoration(labelText: 'Username'),
            ), // TextField
            const SizedBox(height: 10),
            TextField(
              controller: _passwordCtrl,
              obscureText: true,
              decoration: const InputDecoration(labelText: 'Password'),
            ), // TextField
            const SizedBox(height: 30),
            Row(
              mainAxisAlignment: MainAxisAlignment.spaceEvenly,
              children: [
                ElevatedButton(onPressed: _login, child: const Text('Submit')),
                ElevatedButton(
                  onPressed: () {
                    _usernameCtrl.clear();
```

```
                    _passwordCtrl.clear();
                  },
                  style: ElevatedButton.styleFrom(backgroundColor: Colors.grey),
                  child: const Text('Reset'),
                ), // ElevatedButton
              ],
            ), // Row
          ],
        ), // Column
      ), // Padding
    ); // Scaffold
  }
}

// ========================================
// 3. DASHBOARD PRODUK
// ========================================
class DashboardScreen extends StatefulWidget {
  const DashboardScreen({super.key});

  @override
  State<DashboardScreen> createState() => _DashboardScreenState();
}

class _DashboardScreenState extends State<DashboardScreen> {
  double totalJual = 0;

  final List<Map<String, dynamic>> products = [
    {
      'nama': 'COSRX Set',
      'harga': 150000,
      'desc': 'Satu set perawatan wajah COSRX Low pH Good Morning.',
      'gambar': 'assets/Skincare1.jpg',
      'qty': 0,
    },
    {
      'nama': 'Lacoco Serum',
      'harga': 250000,
      'desc': 'Lacoco En Nature Bust Fit Concentrate Serum.',
      'gambar': 'assets/Skincare2.jpg',
      'qty': 0,
    },
```

```
    {
      'nama': 'Tonskin Set',
      'harga': 180000,
      'desc': 'Paket lengkap Tonskin Brightening untuk kulit cerah.',
      'gambar': 'assets/Skincare3.jpg',
      'qty': 0,
    },
    {
      'nama': 'Y.O.U Hydra',
      'harga': 200000,
      'desc': 'Rangkaian skincare Y.O.U HydraRescue Series.',
      'gambar': 'assets/Skincare4.jpg',
      'qty': 0,
    },
    {
      'nama': 'Aqua+ Series',
      'harga': 125000,
      'desc': 'Aqua+ Series HYA 8D PLUS Revitalizing Skindrops.',
      'gambar': 'assets/Skincare5.jpg',
      'qty': 0,
    },
  ];

  Future<void> _launchURL(String urlString) async {
    final Uri url = Uri.parse(urlString);
    if (!await launchUrl(url)) {
      ScaffoldMessenger.of(context).showSnackBar(
        const SnackBar(content: Text('Tidak bisa membuka aksi ini')),
      );
    }
  }

  void _resetKeranjang() {
    setState(() {
      totalJual = 0;
      for (var prod in products) {
        prod['qty'] = 0;
      }
    });
  }

  @override
```

```dart
214   Widget build(BuildContext context) {
215     return Scaffold(
216       appBar: AppBar(
217         title: const Text('Katalog Produk'),
218         actions: [
219           PopupMenuButton<String>(
220             onSelected: (value) {
221               if (value == 'call') _launchURL('tel:081234567890');
222               if (value == 'sms') _launchURL('sms:081234567890');
223               if (value == 'maps') _launchURL('geo:-7.5666,110.8166');
224               if (value == 'update') {
225                 Navigator.push(
226                   context,
227                   MaterialPageRoute(
228                     builder: (context) => const UpdateUserScreen(),
229                   ), // MaterialPageRoute
230                 );
231               }
232             },
233             itemBuilder: (BuildContext context) {
234               return [
235                 const PopupMenuItem(
236                   value: 'call',
237                   child: Text('Call Center Penjual'),
238                 ), // PopupMenuItem
239                 const PopupMenuItem(
240                   value: 'sms',
241                   child: Text('SMS ke Penjual'),
242                 ), // PopupMenuItem
243                 const PopupMenuItem(
244                   value: 'maps',
245                   child: Text('Maps Lokasi Agen'),
246                 ), // PopupMenuItem
247                 const PopupMenuItem(
248                   value: 'update',
249                   child: Text('Update User & Password'),
250                 ), // PopupMenuItem
251               ];
252             },
253           ), // PopupMenuButton
254         ],
255       ), // AppBar
256       body: GridView.builder(
257         padding: const EdgeInsets.all(12),
258         gridDelegate: const SliverGridDelegateWithFixedCrossAxisCount(
```

```
        crossAxisCount: 2,
        childAspectRatio: 0.55,
        crossAxisSpacing: 12,
        mainAxisSpacing: 12,
      ), // SliverGridDelegateWithFixedCrossAxisCount
      itemCount: products.length,
      itemBuilder: (context, index) {
        final prod = products[index];
        return Card(
          elevation: 3,
          shape: RoundedRectangleBorder(
            borderRadius: BorderRadius.circular(12),
          ), // RoundedRectangleBorder
          child: Column(
            crossAxisAlignment: CrossAxisAlignment.stretch,
            children: [
              Expanded(
                child: InkWell(
                  onTap: () {
                    setState(() {
                      prod['qty']++;
                      totalJual += prod['harga'];
                    });
                    ScaffoldMessenger.of(context).showSnackBar(
                      SnackBar(
                        content: Text('${prod['nama']} ditambahkan!'),
                        duration: const Duration(milliseconds: 500),
                      ), // SnackBar
                    );
                  },
                  child: ClipRRect(
                    borderRadius: const BorderRadius.only(
                      topLeft: Radius.circular(12),
                      topRight: Radius.circular(12),
                    ), // BorderRadius.only
                    child: Image.asset(
                      prod['gambar'],
                      fit: BoxFit.cover,
                      errorBuilder: (context, error, stackTrace) {
                        return const Center(
                          child: Icon(
                            Icons.broken_image,
```

```
                        size: 60,
                        color: Colors.grey,
                      ), // Icon
                    ); // Center
                  },
                ), // Image.asset
              ), // ClipRRect
            ), // InkWell
          ), // Expanded
          Padding(
            padding: const EdgeInsets.symmetric(
              horizontal: 8.0,
              vertical: 4.0,
            ), // EdgeInsets.symmetric
            child: Column(
              children: [
                InkWell(
                  onTap: () {
                    showDialog(
                      context: context,
                      builder: (context) => AlertDialog(
                        title: Text(prod['nama']),
                        content: Text(prod['desc']),
                        actions: [
                          TextButton(
                            onPressed: () => Navigator.pop(context),
                            child: const Text('Tutup'),
                          ), // TextButton
                        ],
                      ), // AlertDialog
                    );
                  },
                  child: Text(
                    prod['nama'],
                    style: const TextStyle(
                      fontSize: 15,
                      fontWeight: FontWeight.bold,
                      decoration: TextDecoration.underline,
                    ), // TextStyle
                    maxLines: 1,
                    overflow: TextOverflow.ellipsis,
                  ), // Text
```

```
343                 ), // InkWell
344                 const SizedBox(height: 2),
345                 Text(
346                   'Rp ${prod['harga']}',
347                   style: TextStyle(
348                     fontSize: 14,
349                     color: Colors.teal[800],
350                     fontWeight: FontWeight.w600,
351                   ), // TextStyle
352                 ), // Text
353
354                 Row(
355                   mainAxisAlignment: MainAxisAlignment.center,
356                   children: [
357                     IconButton(
358                       icon: const Icon(
359                         Icons.remove_circle,
360                         color: Colors.red,
361                       ), // Icon
362                       onPressed: () {
363                         if (prod['qty'] > 0) {
364                           setState(() {
365                             prod['qty']--;
366                             totalJual -= prod['harga'];
367                           });
368                         }
369                       },
370                     ), // IconButton
371                     Text(
372                       '${prod['qty']}',
373                       style: const TextStyle(
374                         fontWeight: FontWeight.bold,
375                         fontSize: 16,
376                       ), // TextStyle
377                     ), // Text
378                     IconButton(
379                       icon: const Icon(
380                         Icons.add_circle,
381                         color: Colors.green,
382                       ), // Icon
383                       onPressed: () {
384                         setState(() {
```

```
                                prod['qty']++;
                                totalJual += prod['harga'];
                              });
                            },
                          ), // IconButton
                        ],
                      ), // Row
                    ],
                  ), // Column
                ), // Padding
              ],
            ), // Column
          ); // Card
        },
      ), // GridView.builder
      bottomNavigationBar: InkWell(
        onTap: () {
          if (totalJual > 0) {
            // Ambil daftar barang yang dibeli (qty > 0)
            List<Map<String, dynamic>> keranjang = products
                .where((p) => p['qty'] > 0)
                .toList();

            Navigator.push(
              context,
              MaterialPageRoute(
                builder: (context) =>
                    PaymentScreen(totalBayar: totalJual, keranjang: keranjang),
              ), // MaterialPageRoute
            ).then((value) {
              if (value == true) {
                _resetKeranjang();
              }
            });
          } else {
            ScaffoldMessenger.of(context).showSnackBar(
              const SnackBar(content: Text('Keranjang masih kosong!')),
            );
          }
        },
        child: Container(
          height: 60,
```

```
            color: Colors.teal,
            alignment: Alignment.center,
            child: Text(
              'Total : Rp $totalJual',
              style: const TextStyle(
                color: Colors.white,
                fontSize: 18,
                fontWeight: FontWeight.bold,
              ), // TextStyle
            ), // Text
          ), // Container
        ), // InkWell
      ); // Scaffold
    }
  }

  // ========================================
  // 4. FORM PEMBAYARAN
  // ========================================
  class PaymentScreen extends StatefulWidget {
    final double totalBayar;
    final List<Map<String, dynamic>> keranjang;

    const PaymentScreen({
      super.key,
      required this.totalBayar,
      required this.keranjang,
    });

    @override
    State<PaymentScreen> createState() => _PaymentScreenState();
  }

  class _PaymentScreenState extends State<PaymentScreen> {
    final _bayarCtrl = TextEditingController();
    double kembalian = 0;

    void hitungKembalian() {
      double jmlBayar = double.tryParse(_bayarCtrl.text) ?? 0;
      setState(() {
        kembalian = jmlBayar - widget.totalBayar;
      });
```

```dart
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('Pembayaran')),
      body: SingleChildScrollView(
        padding: const EdgeInsets.all(20.0),
        child: Column(
          crossAxisAlignment: CrossAxisAlignment.stretch,
          children: [
            const Text(
              'Daftar Belanjaan:',
              style: TextStyle(fontSize: 18, fontWeight: FontWeight.bold),
            ), // Text
            const SizedBox(height: 10),
            // ListView untuk menampilkan daftar produk
            ListView.builder(
              shrinkWrap: true,
              physics: const NeverScrollableScrollPhysics(),
              itemCount: widget.keranjang.length,
              itemBuilder: (context, index) {
                final item = widget.keranjang[index];
                final subtotal = item['harga'] * item['qty'];
                return ListTile(
                  contentPadding: EdgeInsets.zero,
                  title: Text(
                    item['nama'],
                    style: const TextStyle(fontWeight: FontWeight.bold),
                  ), // Text
                  subtitle: Text('Rp ${item['harga']} x ${item['qty']}'),
                  trailing: Text(
                    'Rp $subtotal',
                    style: const TextStyle(fontWeight: FontWeight.bold),
                  ), // Text
                ); // ListTile
              },
            ), // ListView.builder
            const Divider(thickness: 2),
            const SizedBox(height: 10),
            Column(
              crossAxisAlignment: CrossAxisAlignment.center,
              children: [
```

```
            const Text(
              'Total Transaksi:',
              style: TextStyle(fontSize: 18, fontWeight: FontWeight.bold),
            ), // Text
            Text(
              'Rp ${widget.totalBayar}',
              style: const TextStyle(
                fontSize: 24,
                color: Colors.teal,
                fontWeight: FontWeight.bold,
              ), // TextStyle
            ), // Text
            const SizedBox(height: 30),
            TextField(
              controller: _bayarCtrl,
              keyboardType: TextInputType.number,
              textAlign: TextAlign.center,
              decoration: const InputDecoration(
                labelText: 'Jumlah Pembayaran (Rp)',
                border: OutlineInputBorder(),
              ), // InputDecoration
            ), // TextField
            const SizedBox(height: 10),
            ElevatedButton(
              onPressed: hitungKembalian,
              child: const Text('Hitung Kembalian'),
            ), // ElevatedButton
            const SizedBox(height: 30),
            const Text(
              'Kembali:',
              style: TextStyle(fontSize: 18, fontWeight: FontWeight.bold),
            ), // Text
            Text(
              'Rp $kembalian',
              style: TextStyle(
                fontSize: 24,
                fontWeight: FontWeight.bold,
                color: kembalian < 0 ? Colors.red : Colors.green,
              ), // TextStyle
            ), // Text
          ],
        ), // Column
```

```
            const SizedBox(height: 40),
            ElevatedButton(
              style: ElevatedButton.styleFrom(
                padding: const EdgeInsets.symmetric(vertical: 15),
              ),
              onPressed: () => Navigator.pop(context, true),
              child: const Text('Selesai', style: TextStyle(fontSize: 18)),
            ), // ElevatedButton
          ],
        ), // Column
      ), // SingleChildScrollView
    ); // Scaffold
  }
}

// ========================================
// 5. UPDATE USER & PASSWORD
// ========================================
class UpdateUserScreen extends StatefulWidget {
  const UpdateUserScreen({super.key});

  @override
  State<UpdateUserScreen> createState() => _UpdateUserScreenState();
}

class _UpdateUserScreenState extends State<UpdateUserScreen> {
  final _userCtrl = TextEditingController();
  final _passCtrl = TextEditingController();

  Future<void> _updateData() async {
    if (_userCtrl.text.isNotEmpty && _passCtrl.text.isNotEmpty) {
      final prefs = await SharedPreferences.getInstance();
      await prefs.setString('username', _userCtrl.text);
      await prefs.setString('password', _passCtrl.text);

      ScaffoldMessenger.of(context).showSnackBar(
        const SnackBar(content: Text('User & Password berhasil diupdate!')),
      );
      Navigator.pop(context);
    }
  }

```

```dart
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('Update User & Password')),
      body: Padding(
        padding: const EdgeInsets.all(20.0),
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            TextField(
              controller: _userCtrl,
              decoration: const InputDecoration(labelText: 'New Username'),
            ), // TextField
            const SizedBox(height: 10),
            TextField(
              controller: _passCtrl,
              obscureText: true,
              decoration: const InputDecoration(labelText: 'New Password'),
            ), // TextField
            const SizedBox(height: 30),
            Row(
              mainAxisAlignment: MainAxisAlignment.spaceEvenly,
              children: [
                ElevatedButton(
                  onPressed: _updateData,
                  child: const Text('Update'),
                ), // ElevatedButton
                ElevatedButton(
                  onPressed: () => Navigator.pop(context),
                  style: ElevatedButton.styleFrom(backgroundColor: Colors.grey),
                  child: const Text('Cancel'),
                ), // ElevatedButton
              ],
            ), // Row
          ],
        ), // Column
      ), // Padding
    ); // Scaffold
  }
}

```